

Spectral Efficiency Analysis of Acoustic OFDM System

1 Introduction

Spectral efficiency (η) quantifies how efficiently a communication system uses the available bandwidth. Is defined as the net data rate (R_{net}) divided by the system bandwidth (B):

$$\eta = \frac{R_{net}}{B} \quad [\text{bit/s/Hz}] \quad (1)$$

To calculate this for the implemented acoustic OFDM system, we must determine the total number of useful bits transmitted and the total time duration of the transmission, taking into account the frame structure and the resampling process.

2 System Parameters

The system is configured with the following parameters:

- **Bandwidth (B):** 2000 Hz (`conf.ofdm.bandwidth`).
- **Sampling Rate (f_s):** 48,000 Hz (`conf.f_s`).
- **Number of Subcarriers (N):** 512 (`conf.ofdm.ncarrier`).
- **Cyclic Prefix (N_{cp}):** 256 samples (`conf.ofdm.cplen`).
- **Modulation Order (M):** QPSK $\rightarrow$ 2 bits/symbol (`conf.modulation_order`).
- **Frame Structure:**
 - Preamble: 500 symbols (Single Carrier).
 - Training: 1 OFDM symbol.
 - Data: 50 OFDM symbols.

3 Theoretical Maximum Spectral Efficiency

The theoretical maximum efficiency (η_{max}) describes the system's performance if we ignore all frame-level overhead (such as the preamble and training symbols) and assume continuous data transmission. It depends solely on the OFDM symbol parameters: the active subcarriers, modulation depth, and guard interval (Cyclic Prefix). The efficiency is the ratio of useful bits per symbol to the time-frequency resources consumed by that symbol:

$$\eta_{max} = \frac{N_{active} \cdot \log_2(M)}{N + N_{cp}} \quad (2)$$

Where:

- N_{active} is the number of data-carrying subcarriers (512 in our case).

- $\log_2(M)$ is the number of bits per subcarrier (2 for QPSK).
- N is the FFT size (512).
- N_{cp} is the Cyclic Prefix length (256).

Substituting the values from the default configuration:

$$\begin{aligned}\eta_{max} &= \frac{512 \times 2}{512 + 256} \\ &= \frac{1024 \text{ bits}}{768 \text{ samples}} \\ &\approx 1.333 \text{ bit/s/Hz}\end{aligned}$$

4 Actual Efficiency Calculation

The total useful bits information is contained only in the data OFDM symbols.

$$\begin{aligned}\text{Total Useful Bits} &= (N \times \log_2 M) \times \text{Number of Data Symbols} \\ &= (512 \times 2) \times 50 = 51,200 \text{ bits}\end{aligned}$$

The system generates signals at baseband (2 kHz) and upsamples them to 48 kHz. We calculate the duration based on the samples sent to the sound card.

4.0.1 A. Preamble Duration

The preamble is 500 symbols at 1000 Hz, upsampled by the transmitter OS factor (48):

$$T_{preamble} = \frac{500 \times 48}{48,000} = 0.5 \text{ s} \quad (3)$$

4.0.2 B. OFDM Block Duration

The OFDM block (Training + 50 Data symbols = 51 symbols) has a baseband length of $51 \times (512 + 256) = 39,168$ samples. The resampling factor is $f_s/B = 24$.

$$T_{OFDM} = \frac{39,168 \times 24}{48,000} = 19.584 \text{ s} \quad (4)$$

So the total duration is:

$$T_{total} = 0.5 + 19.584 = 20.084 \text{ s} \quad (5)$$

Then we can compute the achieved spectral efficiency:

$$\eta = \frac{R_{net}}{B} = \frac{51,200/20.084}{2000} \approx 1.275 \text{ bit/s/Hz} \quad (6)$$

The achieved efficiency calculated in Section 4 is slightly lower due to the time "wasted" on the Preamble and Training Symbol

5 Detailed Improvements and Limitations

Improving spectral efficiency requires modifying system parameters such as the modulation order M , the guard interval N_{cp} , or the ratio of user data to frame overhead. The following analysis explores these potential improvements alongside the physical constraints of the acoustic channel.

5.1 1. Increasing Modulation Order (M)

One of the most direct methods to improve efficiency is upgrading the modulation scheme from QPSK ($M = 4$) to higher orders like 16-QAM ($M = 16$). Since 16-QAM encodes 4 bits per subcarrier compared to 2 bits for QPSK, this modification would double the numerator in the efficiency formula. For the current system configuration, this would raise the theoretical maximum efficiency to:

$$\eta_{max}(16\text{-QAM}) = \frac{512 \times 4}{768} \approx 2.66 \text{ bit/s/Hz}$$

However, this improvement is strictly limited by the available Signal-to-Noise Ratio (SNR). Higher-order constellations pack symbols closer together in the complex plane, making them more susceptible to noise. While QPSK operates reliably at approximately 10–12 dB SNR, 16-QAM typically requires 17–20 dB. In practical acoustic environments, background noise and hardware non-linearities often make it difficult to consistently achieve the high SNR required for dense constellations.

5.2 2. Reducing Cyclic Prefix Length (N_{cp})

The current system utilizes a Cyclic Prefix (CP) of 256 samples, which is 50% of the useful symbol length. Reducing this overhead significantly increases the data rate. For example, shortening the CP to 128 samples would reduce the symbol duration in the denominator, resulting in a theoretical efficiency of:

$$\eta_{max} = \frac{1024}{512 + 128} \approx 1.6 \text{ bit/s/Hz}$$

This represents an approximate 20% increase in efficiency. The limitation of this approach is the risk of Inter-Symbol Interference (ISI). The CP duration must exceed the maximum delay spread of the channel. The current configuration provides a protection interval of $T_{cp} = 128$ ms, which is very conservative for small rooms. However, in larger spaces like lecture halls, significant reverberation could exceed a shortened CP, leading to irreducible bit errors.

5.3 3. Optimizing Frame Structure

Efficiency can also be improved by amortizing the fixed overhead of the synchronization preamble over a larger payload. Currently, the 0.5-second preamble precedes only 50 data symbols. Increasing the frame size to, for instance, 200 data symbols would dilute the preamble's impact on the net data rate. This strategy is constrained by channel time-variance and hardware imperfections. Over longer transmission durations (e.g., exceeding 30 seconds), sampling clock offsets between the transmitter and receiver accumulate, causing progressive phase rotation in the constellation. Furthermore, physical changes in the room (channel aging) may render the initial channel estimate H invalid before the frame ends, necessitating complex continuous pilot tracking.

5.4 4. Adaptive Bit Loading

Finally, the system could employ adaptive bit loading rather than a fixed modulation scheme. By estimating the SNR per subcarrier, the transmitter could assign higher-order modulation (e.g., 16-QAM) to frequencies with constructive interference and robust BPSK to those in deep fades. The primary obstacle to this improvement is complexity and the requirement for a feedback channel. Adaptive loading requires the receiver to analyze the channel state and communicate the optimal bit map back to the transmitter. In a simple one-way acoustic link or a pre-recorded transmission, establishing this real-time feedback loop is not feasible.